DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					
Course Code: 24CS36L	*Course Title: Data Structures Lab*				
Total Credits: 1.5 L-0: T-0: P-45	*SDGs addressed: 4,9,11*	*Course Category: PCCL*	*Course Type: Laboratory*		

Teaching Learning Process		*Total Contact Hours/semester*	*Assessment Weightage and Marks*			
	Lecture	-		CIE	SEE	Total
	Tutorial	-	*Weightage, %*	*40*	*60*	*100*
	Practical	45	*Maximum Marks*	*40*	*60*	*100*
	Self-learning and Term Work	-	*Minimum Marks*	*20*	*25*	*45**
	Total	45				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: C programming.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Implement programs to illustrate the use of pointers to arrays, structures and functions.	L3	WK2, WK6
CO2	Implement various operations on Linear lists with Array and Linked list representations and analyze the performance of operations.	L3	WK2, WK4
CO3	Implement different data structures such as Stack, Queue, Tree and Graph using Array and Linked list representations with static or dynamic memory allocation and analyze the performance of operations, use stack for expression conversion and evaluation.	L3	WK2, WK4

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

SI No	Programs
1.	Define a structure called Student with the members: Name, Reg_no, marks in 3 tests and average_ marks. Develop a menu driven program to perform the following by writing separate function for each operation: a) read information of N students b) display student's information c) to calculate the average of best two test marks of each student. d) Sort students based on Reg_no. Note: Allocate memory dynamically and illustrate the use of pointer to an array of structure.
2.	Develop a menu driven program to implement the following operations on an array of integers with dynamic memory allocation. Display the array contents after each operation. i)Insert by position ii) Delete by key iii) Search by position iv) Reverse the contents.
3.	Develop a menu driven program to implement the following operations on an array of integers with dynamic memory allocation. Display the array contents after each operation. i) Insert by order ii) Delete by position iii) Search by key iv) Reverse the contents.
4.	Implement single (circular or non-circular) linked list to perform the following operations

	i) Insert by order ii) Delete at beginning iii) Search for an item by key iv) Delete by key Display the list contents after each operation.
5.	Implement circular double linked list with header to perform the following operations i) Insert by order ii) Delete by position iii) Delete by key iv) Search by position Display the list contents after each operation.
6.	Implement single (circular or non-circular) linked list to perform the following operations i)Insert front ii) Insert at position iii) Delete a node with the given key iv) Search for an item by position. Display the list contents after each operation.
7.	Implement circular double linked list with header to perform the following operations i)Insert front ii) Insert at position iii) Delete by position iv) Search by key Display the list contents after each operation.
8.	Implement double (circular or non-circular) linked list to perform the following operations. i)Insert by position ii) Delete rear iii) Delete at position iv) Search for an item by value Display the list contents after each operation.
9.	Implement circular single linked list with header to perform the following operations i)Insert by order ii) Delete rear iii) Delete Front iv) reverse operation Display the list contents after each operation.
10.	Develop a menu driven program to convert infix expression to postfix expression using stack and evaluate the postfix expression. (Test for nested parenthesized expressions). **Note: Define Stack structure and implement the operations. Use different stacks for conversion and evaluation.**
11.	Develop a menu driven program to convert infix expression to prefix expression using stack and evaluate the prefix expression (Test for nested parenthesized expressions). **Note: Define Stack structure and implement the operations. Use different stacks for conversion and evaluation.**
12.	Develop a menu driven program to implement the following types of Queues with array representation by allocating memory dynamically. i)Ordinary Queue ii) Double ended Queue **Note: Define Queue structure and implement the operations.**
13.	Develop a menu driven program to implement the following types of Queues with array representation by allocating memory dynamically. i) Circular Queue ii) Priority Queue **Note: Define Queue structure and implement the operations.**
14.	Develop a menu driven program to implement Binary Search tree with the following operations. i)Construction ii) Traversals (DFS and BFS) iii) Finding node with the Maximum key value and printing the node details along with the parent. iv) Counting all nodes. v) Finding height.
15.	Develop a menu driven program to implement Binary Search tree with the following operations. i) Construction ii) Traversals (DFS and BFS) iii) Searching a node by key and deleting if exists (node to be deleted may be leaf or non- leaf with one child or two children)
16.	Develop a menu driven program to implement Graph traversal techniques.